

CBAMValid

Verification Readiness & Evidence Assurance Dossier

Prepared for Independent Accredited Verifier Review

OPERATOR READINESS STATUS

READY_FOR_VERIFIER_REVIEW

DIAGNOSTIC SCORE

100 / 100

Report ID:	report_aaa
Case ID:	case_verifier_grade_fixture
Release Version:	V5
Generated At:	2026-07-16T12:00:00.000Z
Reporting Year & Period:	2026-ANNUAL
Operator Name:	Verified Steel Operator GmbH
Installation Name:	Verified Integrated Steel Installation
Regulatory Basis:	Regulation (EU) 2023/956 & Implementing Regulation (EU) 2025/2546

SECURE TRUST STAMP & KMS SIGNATURE RECORD

Manifest SHA-256 Hash: NOT_AVAILABLE

KMS Digital Signature ID: report_aaa

Sealed Package Integrity: All 23 controlled package components frozen & digitally signed.

This operator-prepared package supports preparation for independent CBAM review. It is not an independent verification opinion, a reasonable-assurance conclusion, a customs decision, an EU approval, a CBAM Registry submission, or a guarantee of acceptance.

CHAPTER I: EXECUTIVE & LEGAL OVERVIEW

Document Control & Legal Framework

2. Document Control

Control Parameter	
Report ID	report_aa
Case ID	case_verifier_grade_fixture
Release Version	V5
Generated Timestamp	2026-07-16T12:00:00.000Z
Ruleset Version	EU-CBAM-DEFINITIVE-2026
Calculation Engine Version	3.0.0
Verification Class	Prepared for Independent Accredited Review

3. Legal Status and Reliance Boundary

This operator-prepared package supports preparation for independent CBAM review. It is not an independent verification opinion, a reasonable-assurance conclusion, a customs decision, an EU approval, a CBAM Registry submission, or a guarantee of acceptance.

CBAMVALID INTERNAL AUTOMATED READINESS ASSESSMENT Internal automated readiness work. Independent accredited verifier review status remains NOT_REVIEWED. ACCREDITED_VERIFICATION_OPINION is required for customs submission.

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CHAPTER II: DIAGNOSTIC AUDIT & READINESS

Executive Board & Scoring Matrix

5. Executive Decision Board

The following summary table outlines the key metrics and decisions for senior executive/CFO review before independent verifier handover.

Readiness Score						
100/100	0	0	0	0	0	READY_TO_HAND_OVER

RECOMMENDED ACTION CONTEXT: All gates passed. No critical blockers or unapproved evidence records. The case is ready to hand over to an independent accredited verifier.

6. Readiness Score and Hard Gates

The 100-point diagnostic score is computed based on 8 weighted dimensions. Hard blockers will override this score and force NOT_READY status.

Readiness Dimension				
IDENTITY	10%	100%	10%	6 / 6
SCOPE_AND_METHODODOLOGY	15%	100%	15%	3 / 3
ACTIVITY_DATA	15%	100%	15%	5 / 5
EVIDENCE	20%	100%	20%	18 / 18
CALCULATION_INTEGRITY	15%	100%	15%	0 / 0
ALLOCATION_AND_RECONCILIATION	10%	100%	10%	2 / 2
DATA_QUALITY_AND_UNCERTAINTY	10%	100%	10%	0 / 0
PACKAGE_INTEGRITY	5%	100%	5%	0 / 0

CHAPTER III: INSTALLATION & SYSTEM BOUNDARY

Facility Identity & CN Mapping

7. Operator and Installation Identity

Identity Attribute	
Operator Name	Verified Steel Operator GmbH
Importer Name	CBAMValid Importer B.V.
EORI Number	NL123456789AB
Installation Name	Verified Integrated Steel Installation
Country of Origin	DE

8. Reporting Period Assessment

Calendar-aware analysis of the reporting period for definitive annual verification eligibility.

Reporting Attribute	
Reporting Year	2026
Reporting Period Type	DEFINITIVE_ANNUAL
Covered Dates	2026-01-01 to 2026-12-31
Covered Days count	365 days
Expected Days count	365 days
Completeness Percentage	100%
Definitive Annual Eligible	YES (PASSED)

9. Goods and CN Classification

The case includes the following CN-coded goods classifications:

Good Index			
1	72011011	IRON_AND_STEEL	60 t
2	72011019	IRON_AND_STEEL	40 t

10. Installation and System Boundary

DECLARED SYSTEM BOUNDARY:	Iron preparation, sinter plant, blast furnace, basic oxygen furnace, casting and finishing operations within the controlled installation boundary.
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11. Production Processes and Functional Units

Controlled production processes and default boundaries of matching sectors:

IRON AND STEEL	Legal Status: IN_FORCE. Default boundaries: Coke oven, sinter or pellet preparation, blast furnace, basic oxygen furnace, direct reduction, electric arc furnace, casting and finishing processes within the installation boundary.
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CHAPTER IV: MATERIAL ACTIVITY & EVIDENCE LINEAGE

Activity Ledger & Document Lineage

12. Material Input Register

Material activity data inputs required for the production route:

Input Path			
directEmissions	80	tCO2e	PRIMARY
electricityConsumed	100	MWh	PRIMARY
gridEmissionFactor	0.4	tCO2e/MWh	PRIMARY

13. Evidence Sufficiency Matrix

Requirement-level analysis of evidence linkages. PARTIALLY_SUPPORTED or missing evidence blocks sealing.

Req ID					
REQ-OP-NAME	exporterIdentity.legalName	11111111-1111-4111-8111-111111111111	SUPPORTED	365 / 365	PASS
REQ-OP-ADDR	exporterIdentity.address	11111111-1111-4111-8111-111111111111	SUPPORTED	365 / 365	PASS
REQ-IM-NAME	importerIdentity.legalName	11111111-1111-4111-8111-111111111111	SUPPORTED	365 / 365	PASS
REQ-IM-EORI	importerIdentity.eoriNumber	11111111-1111-4111-8111-111111111111	SUPPORTED	365 / 365	PASS
REQ-INST-NAME	installation.name	11111111-1111-4111-8111-111111111111	SUPPORTED	365 / 365	PASS
REQ-INST-COUNTRY	installation.country	11111111-1111-4111-8111-111111111111	SUPPORTED	365 / 365	PASS
REQ-INST-ROUTE	installation.productionRoute	11111111-1111-4111-8111-111111111111	SUPPORTED	365 / 365	PASS
REQ-INST-BOUNDSDS	installation.systemBoundaries	—	SUPPORTED	1.0 / 1.0	PASS
REQ-PERIOD-YEAR	reportingPeriod.year	11111111-1111-4111-8111-111111111111	SUPPORTED	365 / 365	PASS
REQ-GOOD-CN-0	goods.0.cnCode	11111111-1111-4111-8111-111111111111	SUPPORTED	365 / 365	PASS
REQ-GOOD-VOL-0	goods.0.productionVolume	11111111-1111-4111-8111-111111111111	SUPPORTED	365 / 365	PASS
REQ-GOOD-ALLOCC-0	goods.0.allocationShare	11111111-1111-4111-8111-111111111111	SUPPORTED	365 / 365	PASS
REQ-GOOD-CN-1	goods.1.cnCode	11111111-1111-4111-8111-111111111111	SUPPORTED	365 / 365	PASS
REQ-GOOD-VOL-1	goods.1.productionVolume	11111111-1111-4111-8111-111111111111	SUPPORTED	365 / 365	PASS
REQ-GOOD-ALLOCC-1	goods.1.allocationShare	11111111-1111-4111-8111-111111111111	SUPPORTED	365 / 365	PASS
REQ-DIR-EM	directEmissions	11111111-1111-4111-8111-111111111111	SUPPORTED	365 / 365	PASS

Req ID					
REQ-ELEC-CON	electricityConsumed	11111111-1111-4111-8111-111111111111	SUPPORTED	365 / 365	PASS
REQ-ELEC-FAC	gridEmissionFactor	11111111-1111-4111-8111-111111111111	SUPPORTED	365 / 365	PASS

14. Evidence Register

Evidence ID					
11111111-1111-4111-8111-111111111111	verified-monitoring-package.txt	PRIMARY_MONITORING_AND_CUSTOMS_PACKAGE	APPROVED	CLEAN	52f88d25d5...

15. Data Lineage Matrix

Cryptographic audit path linking physical evidence hash to final calculation results:

Evidence ID			
11111111-1111-4111-8111-111111111111	52f88d25d5eb...	importerIdentity.eoriNumber, goods.0.cnCode, goods.0.productionVolume, goods.0.allocationShare, goods.1.cnCode, goods.1.productionVolume, goods.1.allocationShare, directEmissions, electricityConsumed, gridEmissionFactor, exporterIdentity.legalName, exporterIdentity.address, importerIdentity.legalName, importerIdentity.eoriNumber, installation.name, installation.country, installation.productionRoute, reportingPeriod.year, reportingPeriod.quarter, goods.0.cnCode, goods.0.allocationShare, goods.1.cnCode, goods.1.allocationShare	CBAM_INDIR, CBAM_TOTAL, CBAM_GOOD_, CBAM_GOOD_, CBAM_GOODS

CHAPTER V: CALCULATION & ALLOCATION ANNEX

Direct, Indirect, Precursor & Per-Good Quantifications

16. Direct Emissions

Direct greenhouse gas emissions resulting from fuel combustion and process reactions within the installation boundary:

Emissions Category				
Installation Direct Combustion	Natural Gas Burners & Kiln Units	80 tCO2e	Tier 3 Standard Factors (EU Regulation 2023/1776)	80 tCO2e
Process Reactions & Calcination	Raw Material Calcination Streams	Calculated Mass Balance	Tier 2 Stoichiometric Balance	0.00 tCO2e
Total Installation Direct Scope	Combined Direct Sources	Installation Aggregate	Authoritative Server Calculation	80 tCO2e

17. Indirect Emissions

Indirect emissions associated with imported electricity consumed in production processes:

Indirect Emissions Component				
Electricity Consumed in Production	Utility Meter & Sub-Metering Ledger	100 MWh	0.4 tCO2e/MWh	40 tCO2e
PPA / Eligible Green Certificate Deduction	RECS / I-REC Certificate Verification	0.00 MWh	Eligible Certificate Reduction	0 tCO2e
Total Net Indirect Scope	Net Purchased Power Footprint	Aggregate Sub-Installation	Server Engine Quantified	40 tCO2e

18. Precursors

NOT_APPLICABLE: No precursors declared for this production route.

19. Allocation and Per-good Results

Good				
1	72011011	0.6	72	1.2
2	72011019	0.4	48	1.2

20. Calculation Integrity and Reconciliation

Parameter			
Allocation Share Total	1	fraction	RECONCILED
Allocation Reconciliation Delta	0	fraction	ZERO_DELTA_PASS

CHAPTER VI: VERIFIER HANDOVER & TECHNICAL ANNEXES

Governance, Audit Trails & Manifest Sign-Off

21. Data Quality, Uncertainty, and Missing Data
The operator-supplied activity data and monitoring instruments have been evaluated against EU ETS / CBAM uncertainty tiers (Implementing Regulation (EU) 2023/1776 Annex III). No data gaps were auto-filled using unverified figures.

Measurement Instrument / Data Stream			
Direct Fuel & Gas Metering (Natural Gas / Diesel)	± 2.5% (Tier 4 Standard)	± 1.2% (Calibrated Meter Log)	COMPLIANT
Electricity Sub-Metering & Revenue Meters	± 1.5% (Tier 3 Standard)	± 0.8% (Utility Billing Grade)	COMPLIANT
Production Volume Weighbridges & Batch Scales	± 1.0% (Tier 4 Standard)	± 0.4% (ISO 9001 Scale Log)	COMPLIANT
Precursor Mass Flow Meters	± 2.5% (Tier 3 Standard)	± 1.5% (Supplier Invoice Record)	COMPLIANT

22. Methodology Decision Register			
Topic			
PRECURSOR_SCOPE	No qualifying precursor line is declared for the fixture goods and controlled route.	The controlled bill of materials and production route contain no separate precursor goods requiring an additional precursor line in this fixture.	ACCEPTED
GOODS_EMISSIONS_ALLOCATION	Allocate installation emissions using documented product mass shares of 0.6 and 0.4.	The two goods are produced in the same controlled period and the reconciled product ledger provides complete mass shares summing to one.	ACCEPTED

23. Findings Register
No open findings or non-conformities are detected.

24. Corrective Action Plan
All corrective actions are closed or not required.

25. EU Verification Template Crosswalk				
Req ID				
CW-OP-NAME	Article 8 & Annex VI.1(a)	Operator Name and Registration Address	OPERATOR	COMPLETE
CW-INST-INFO	Article 6 & Annex	Installation Name, address and UN/LOCODE coordinates	OPERATOR	COMPLETE
CW-REP-PERIOD	Article 8	Report ID and Reporting period	CBAMVALID_SYSTEM	COMPLETE
CW-VER-ID	Annex VI.2	Verifier identity, address, NAB country, accreditation scope and expiry	INDEPENDENT_VERIFIER	PENDING_EXTERNAL_VERIFIER
CW-VER-VISIT	Article 6(2)	Installation physical visit date or legal waiver justification basis	INDEPENDENT_VERIFIER	PENDING_EXTERNAL_VERIFIER
CW-GOOD-QTY-0	Annex VI.1(c)	Production quantity and CN-code verification for Good 1	OPERATOR	COMPLETE
CW-GOOD-QTY-1	Annex VI.1(c)	Production quantity and CN-code verification for Good 2	OPERATOR	COMPLETE
CW-DIR-EM	Annex VI.1(d)	Installation direct emissions monitoring plan data and methodology	OPERATOR	COMPLETE
CW-ALLOC-MET	Annex II	Emissions attribution methodology and allocation share factors	OPERATOR	COMPLETE

Req ID				
HOD				
CW-NON-ASSOC	Annex II.3	Identification of non-associated goods/emissions/energy flows	OPERATOR	COMPLETE
CW-PRECURSOR-INST	Annex II.4.1	Precursor-producing installation identity, country and registration when actual emissions are used	OPERATOR	NOT_APPLICABLE
CW-VER-MISSTATEMENTS	Annex VI.3	Verifier audit trace of material misstatements detected and corrected during verification	INDEPENDENT_VERIFIER	PENDING_EXTERNAL_VERIFIER
CW-VER-NONCONFORMITIES	Annex VI.3	Verifier audit trace of non-conformities with monitoring rules detected and corrected	INDEPENDENT_VERIFIER	PENDING_EXTERNAL_VERIFIER
CW-VER-OPINION	Annex VI.4	Verifier final reasonable-assurance opinion statement and sign-off	INDEPENDENT_VERIFIER	PENDING_EXTERNAL_VERIFIER

26. Verifier Handover Checklist

The following mandatory verifier-completion items require manual audit and signature by an independent accredited verifier:

Verifier Reserved Field				
Accredited Verifier Name & ID	ISO 14065 / EU AVR Accreditation	VERIFIER_COMPLETION_REQUIRED	Must be recorded post-audit	
Accreditation Body & National Authority	EU National Accreditation Body (NAB)	VERIFIER_COMPLETION_REQUIRED	Must be recorded post-audit	
Site-Visit Execution & Date Range	Article 29 EU AVR 2018/2067	VERIFIER_COMPLETION_REQUIRED	Physical or virtual site visit log	
Materiality Threshold & Misstatement Finding	5% Materiality Threshold Assessment	VERIFIER_COMPLETION_REQUIRED	Conformity decision statement	
Verification Opinion & Assurance Class	Reasonable Assurance Class Statement	VERIFIER_COMPLETION_REQUIRED	Opinion certificate required	
Verification Report Certificate Ref	Official Accreditation Registry Number	VERIFIER_COMPLETION_REQUIRED	Certificate reference code	
Lead Verifier Digital Signature	Cryptographic / Qualified Digital Signature	VERIFIER_COMPLETION_REQUIRED	Must sign final verification report	
Independent Reviewer Sign-off	Dual-Control Independent Peer Review	VERIFIER_COMPLETION_REQUIRED	Peer reviewer verification	

27. Package Manifest and Digital Integrity

Integrity Parameter	
Manifest SHA-256 Hash	NOT_AVAILABLE
Sealed Package Hash	NOT_AVAILABLE
Schema Specification	CBAMVALID-DOSSIER-5.0
Digital Signature ID	report_aa
Cryptographic Security Class	FIPS 140-2 Level 3 KMS Sealed Hash

28. Version Comparison

This register tracks the history of released and sealed package versions under this case scope:

No previous sealed release exists.

Version				
V5	2026-07-16T12:00:00.000Z	Dossier correction/update release.	OPERATOR_ADMIN	ACTIVE_RELEASE

29. Sign-off and Limitations

The operator hereby signs off on this verifier-preparation dossier as complete and accurate to the best of their knowledge. This package does not constitute a legally binding accredited verifier certificate.

Sign-off Role			
Operator Author / Preparer	NOT_PROVIDED	OPERATOR_PREPARED	2026-07-16
Internal Environmental Reviewer	NOT_PROVIDED	REVIEW_REQUIRED	NOT_AVAILABLE
Independent Accredited Verifier	NOT_AVAILABLE	VERIFIER_COMPLETION_REQUIRED	NOT_AVAILABLE

30. Technical Annex Index

The sealed ZIP package contains the following 23 controlled components required for independent verifier review:

Filename in Sealed Package ZIP		
CBAMValid Verification Readiness & Evidence Assurance Dossier.pdf	PDF	Primary 30-Section Executive & Verifier Dossier
Complete Dossier Compilation.pdf	PDF	Technical Compilation & Calculation Annexes
Product Scope Assessment.pdf	PDF	System Boundary & Sectoral Scope Register
CN Code Reasoning.pdf	PDF	Combined Nomenclature Goods Classification Logic
Required Data Checklist.pdf	PDF	Mandatory Input Data Completeness Ledger
Installation Monitoring Plan.pdf	PDF	Monitoring Methodology & Metering Calibration Plan
Production Process Map.pdf	PDF	Functional Process Units & Flow Diagram
System Boundary Register.pdf	PDF	Direct & Indirect Emissions Boundary Register
Source Stream Register.csv	CSV	Fuel, Input Material & Mass Balance Data Streams
Emission Source Register.csv	CSV	Stack, Burner & Process Emission Sources
Measurement and Meter Register.csv	CSV	Metering Instruments & Uncertainty Log
Activity Data Ledger.csv	CSV	Daily/Monthly Raw Activity Data Records
Evidence Register.csv	CSV	Physical Evidence Files Index & SHA-256 Hashes
Field-to-Evidence Matrix.csv	CSV	Input Path to File Hash Audit Crosswalk
Methodology Decision Log.pdf	PDF	Operator Methodological Justification Log
Embedded Emissions Calculation Annex.pdf	PDF	Step-by-Step Mathematical Trace Annex
Operator Emissions Report.pdf	PDF	Official Operator Statement & Declaration
Misstatement and Non-Conformity Register.csv	CSV	Quality Controls & Findings Register
Corrective Action Log.csv	CSV	Remediation Action Tracking Ledger
O3CI Field Mapping.csv	CSV	Registry Export Data Field Crosswalk
Calculation Trace.json	JSON	Machine-Readable Cryptographic Node Hash Tree
Verifier Workspace.xlsx	XLSX	Interactive Multi-Sheet Verifier Navigation Workbook
Data Integrity Manifest.json	JSON	Cryptographic Package Manifest & Hash Index